

# Genetic risk accentuates dietary effects on hepatic steatosis, inflammation and fibrosis in a population-based cohort

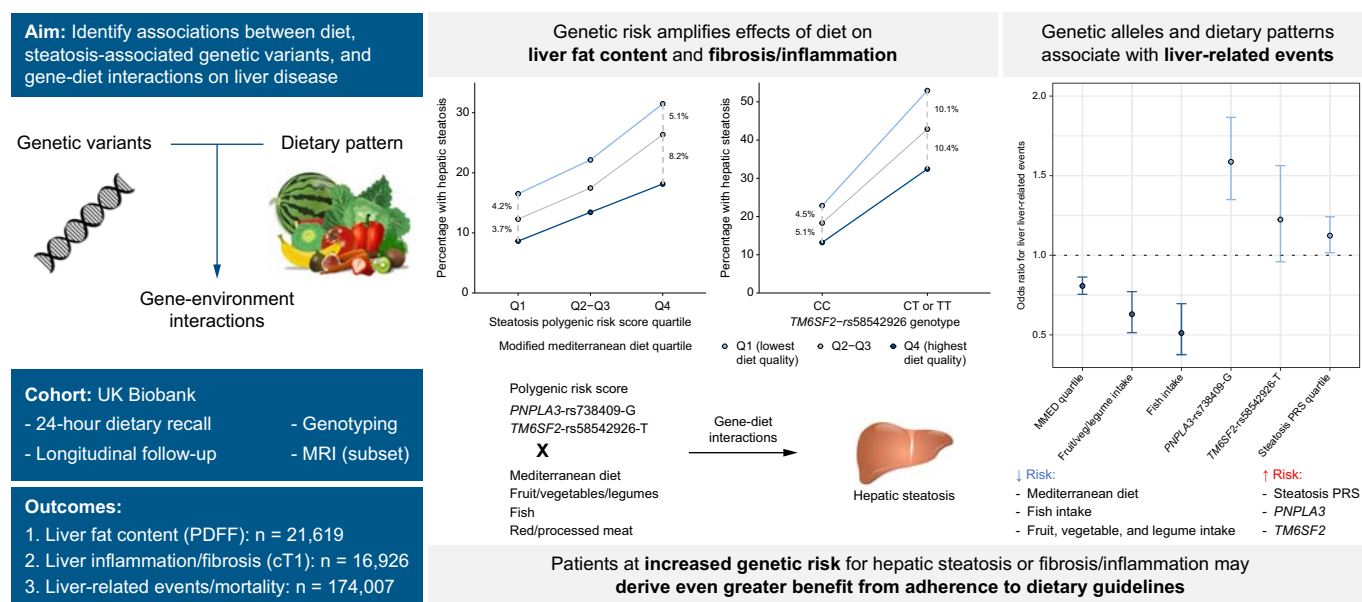
## Authors

Vincent L. Chen, Xiaomeng Du, Antonino Oliveri, ..., Brian D. Halligan, Michael A. Province, Elizabeth K. Speliotes

## Correspondence

espeliot@med.umich.edu (E.K. Speliotes).

## Graphical abstract



## Highlights

- Poor diet quality and specific genetic alleles are associated with steatotic liver disease.
- Diet has a much greater impact on hepatic steatosis in those at higher genetic risk.
- These effects were seen for both liver fat content and iron-controlled T1 time (a measure of fibrosis and inflammation).
- Hence, dietary interventions may have a greater impact in populations at higher genetic risk.

## Impact and implications

Genetic variants and diet both influence risk of hepatic steatosis, inflammation/fibrosis, and hepatic decompensation; however, how gene-diet interactions influence these outcomes has previously not been comprehensively characterized. We investigated this topic in the community-based UK Biobank and found that genetic risk and dietary quality interacted to influence hepatic steatosis and inflammation/fibrosis on liver MRI, so that the effects of diet were greater in people at elevated genetic risk. These results are relevant for patients and medical providers because they show that genetic risk is not fixed (*i.e.* modifiable factors can mitigate or exacerbate this risk) and realistic dietary changes may result in meaningful improvement in liver steatosis and inflammation/fibrosis. As genotyping becomes more routinely used in clinical practice, patients identified to be at high baseline genetic risk may benefit even more from intensive dietary counseling than those at lower risk, though future prospective studies are required.

# Targeting EFHD2 inhibits interferon- $\gamma$ signaling and ameliorates non-alcoholic steatohepatitis

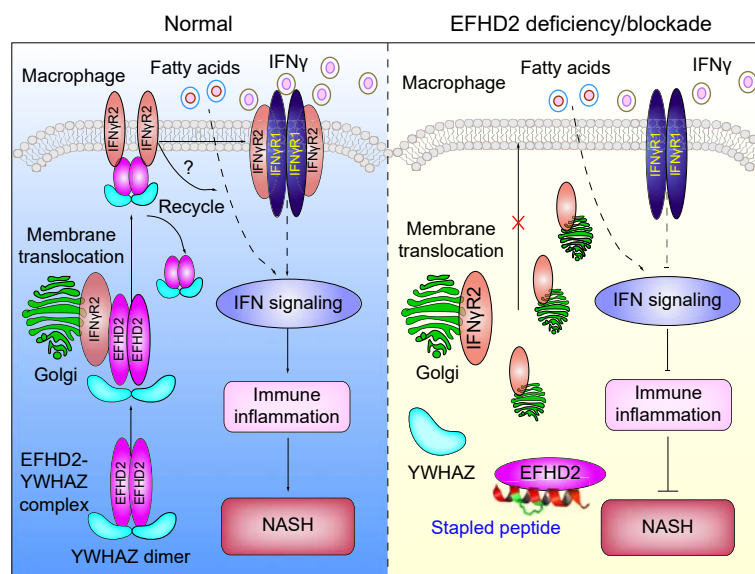
## Authors

Jiang-Tao Fu, Jian Liu, Wen-Bin Wu, ..., Xiang Li, Zhi-Bin Wang, Pei Wang

## Correspondence

pwang@smmu.edu.cn (P. Wang), methyl@smmu.edu.cn (Z.-B. Wang), xiangli@smmu.edu.cn (X. Li).

## Graphical abstract



## Highlights

- EFHD2 exhibits specific expression in hepatic macrophages/monocytes and is induced in NASH liver.
- Global or myeloid cell-specific deletion of EFHD2 improves NASH and hinders NASH-HCC progression.
- EFHD2 regulates the replacement of resident Kupffer cells by infiltrating monocytes during NASH.
- EFHD2 interacts with YWHAZ, promoting translocation of IFN $\gamma$ R2 to the plasma membrane and, in turn, mediating IFN $\gamma$  signaling.
- A novel stapled peptide designed to target EFHD2 demonstrates protective effects against NASH in mice.

## Impact and implications

Non-alcoholic steatohepatitis (NASH) represents an advanced stage of non-alcoholic fatty liver disease (NAFLD); however, not all patients with NAFLD progress to NASH. A key challenge is identifying the factors that trigger inflammation, which propels the transition from simple fatty liver to NASH. Our research pinpointed EFHD2 as a pivotal driver of NASH, orchestrating the over-activation of interferon- $\gamma$  signaling within the liver during NASH progression. A stapled peptide designed to target EFHD2 exhibited therapeutic promise in NASH mice. These findings support the potential of EFHD2 as a therapeutic target in NASH.

# JNJ-73763989 and bersacapavir treatment in nucleos(t)ide analogue-suppressed patients with chronic hepatitis B: REEF-2

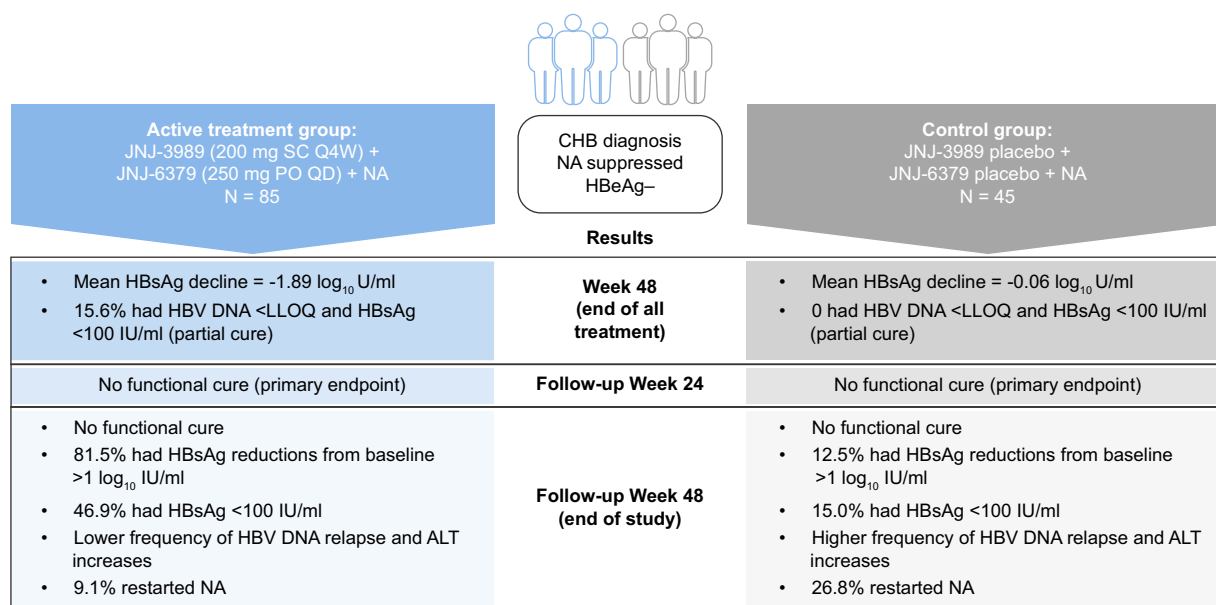
## Authors

Kosh Agarwal, Maria Buti, Florian van Bömmel, ..., Ronald Kalmeijer, Michael Biermer, Isabelle Lonjon-Domanec

## Correspondence

kosh.agarwal@nhs.net (K. Agarwal).

## Graphical abstract



## Highlights

- No functional cure achieved 24 or 48 weeks after 48 weeks of JNJ-3989 + JNJ-6379 + NA.
- All patients had pronounced declines in HBsAg after 48 weeks of JNJ-3989 + JNJ-6379 + NA.
- Maintained HBsAg declines from JNJ-3989 + JNJ-6379 + NA up to 48-weeks off-treatment.
- Pronounced HBV DNA and HBsAg suppression off-treatment after JNJ-3989 + JNJ-6379 + NA.
- JNJ-3989 + JNJ-6379 treatment was safe and well tolerated.

## Impact and implications

Achieving a functional cure from chronic hepatitis B (CHB) with finite treatments is a major unmet medical need. The current study assessed the rate of functional cure and clinical outcome after controlled nucleos(t)ide analogue (NA) withdrawal in patients with low levels of HBsAg induced by 48 weeks of treatment with the small-interfering RNA JNJ-3989 and the capsid assembly modulator JNJ-6379 plus NA vs. patients who only received NA treatment. Though functional cure was not achieved by any patient in either arm, the 48-week treatment regimen of JNJ-3989, JNJ-6379, and NA did result in more patients achieving pronounced reductions in HBsAg, with clinically meaningful reductions maintained for up to 48 weeks off all treatments, as well as fewer off-treatment HBV DNA increases and alanine aminotransferase flares. These findings provide valuable insights for future studies investigating potential finite treatment options, while the reported efficacy and safety outcomes may be of interest to healthcare providers making treatment decisions for patients with NA-suppressed HBeAg-negative CHB.

# B-cell characteristics define HCV reinfection outcome

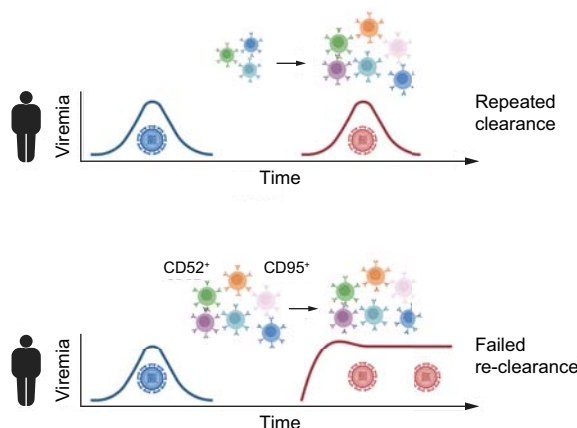
## Authors

Alexander P. Underwood, Money Gupta, Bing-Ru Wu, ..., Heidi E. Drummer, Andrew R. Lloyd, Rowena A. Bull

## Correspondence

r.bull@unsw.edu.au (R.A. Bull).

## Graphical abstract



## Highlights

- Protection from persistent reinfection was associated with expansion of E2-specific B cells upon reinfection.
- E2-specific B cells have a transcriptomic profile associated with activation and proliferation prior to clearance of reinfection.
- We observed evidence of antigen imprinting in HCV reinfection.

## Impact and implications

HCV continues to have a major health burden globally. Limitations in the health infrastructure for diagnosis and treatment, as well as high rates of reinfection, indicate that a vaccine that can protect against chronic HCV infection will greatly complement current efforts to eliminate HCV-related disease. With alternative approaches to testing vaccines, such as controlled human inoculation trials under consideration, we desperately need to identify the correlates of immune protection. In this study, in a small but rare cohort of high-risk injecting drug users who were reinfected multiple times, breadth of neutralisation was not associated with ultimate clearance of the reinfection event. Alternatively, characteristics of the HCV-specific B-cell response associated with B-cell proliferation were. This study indicates that humoral responses are important for protection and suggests that for genetically very diverse viruses, such as HCV, it may be beneficial to look beyond just antibodies as correlates of protection.

# Prominent role of gut dysbiosis in the pathogenesis of cystic fibrosis-related liver disease in mice

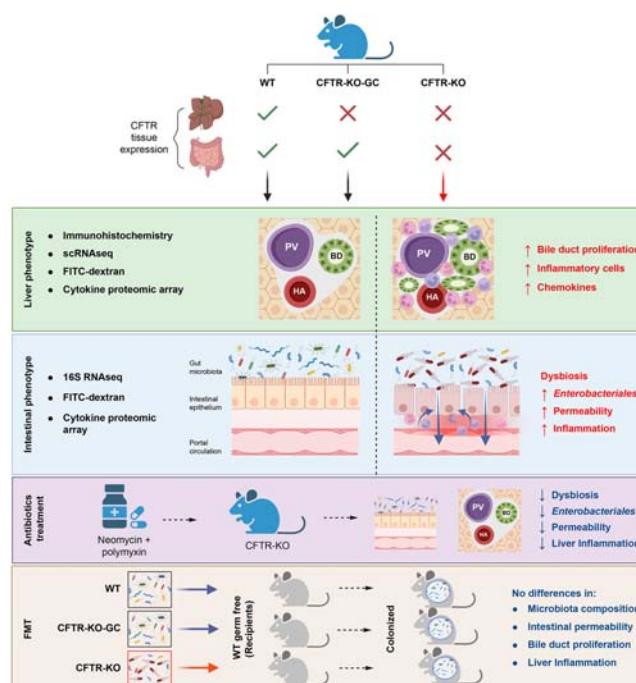
## Authors

Anna Bertolini, Mytien Nguyen, Syeda Andleeb Zehra, ..., Noah Palm, Mario Strazzabosco, Romina Fiorotto

## Correspondence

romina.fiorotto@yale.edu (R. Fiorotto).

## Graphical abstract



## Highlights

- The CF gut microbiota features reduced diversity and increased Enterobacteriales.
- Gut dysbiosis in CF is associated with gut inflammation and increased permeability.
- Intestinal CFTR deficiency is key to the development of liver disease in CF mice.
- Restoring intestinal CFTR or modulating microbiota may improve liver disease in CF.

## Impact and implications

Severe cystic fibrosis-related liver disease (CFLD) affects 10% of patients with cystic fibrosis (CF) and contributes to increased morbidity and mortality. Treatment options remain limited due to a lack of understanding of disease pathophysiology. The cystic fibrosis transmembrane conductance regulator (CFTR) mediates  $\text{Cl}^-$  and  $\text{HCO}_3^-$  secretion in the biliary epithelium and its defective function is thought to cause cholestasis and excessive inflammatory responses in CF. However, our study in *Cftr*-knockout mice demonstrates that microbial dysbiosis, combined with increased intestinal permeability caused by defective CFTR in the intestinal mucosa, acts as a necessary co-factor for the development of CFLD-like liver pathology in mice. These findings uncover a major role for the gut microbiota in CFLD pathogenesis and call for further investigation and clinical validation to develop targeted therapeutic strategies acting on the gut-liver axis in CF.

# Prospective validation of the EASL management algorithm for acute kidney injury in cirrhosis

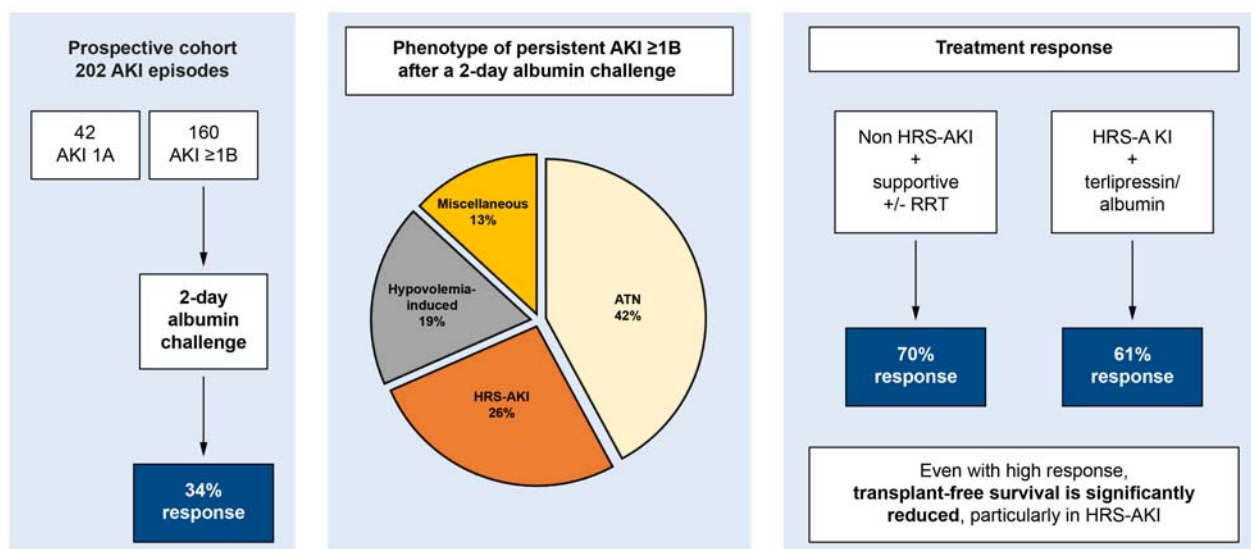
## Authors

Ann Thu Ma, Cristina Solé, Adrià Juanola, ..., Isabel Graupera, Elisa Pose, Pere Ginès

## Correspondence

pgines@clinic.cat (P. Ginès).

## Graphical abstract



## Highlights

- In 2018, EASL published an algorithm for the diagnosis/management of acute kidney injury in patients with cirrhosis.
- This algorithm had not been validated in real-world practice.
- In this prospective study, this algorithm was associated with high renal response rates, both overall and in different phenotypes.
- The use of the algorithm resulted in the swift diagnosis and treatment of hepatorenal syndrome.
- These results support the use of this algorithm in clinical practice.

## Impact and implications

The occurrence of acute kidney injury (AKI) in patients with cirrhosis is associated with poor short-term mortality. Improving its rapid identification and prompt management was the focus of the recently proposed EASL AKI algorithm. This is the first prospective study demonstrating that high AKI response rates are achieved with the use of this algorithm, which includes identification of AKI, treatment of precipitating factors, a 2-day albumin challenge in patients with AKI ≥1B, and supportive therapy in patients with persistent AKI not meeting HRS-AKI criteria or terlipressin with albumin in those with HRS-AKI. These findings support the use of this algorithm in clinical practice.



# Recurrent splanchnic and extrasplanchnic thrombotic events in patients with non-cirrhotic portal vein thrombosis associated with local factors

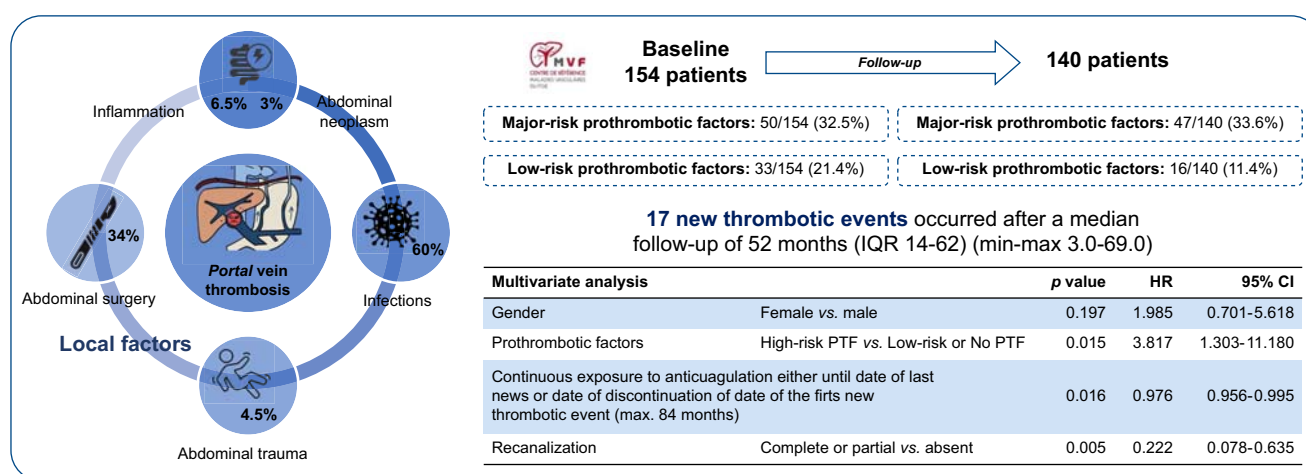
## Authors

Isabelle Ollivier-Hourmand, Louise Lebedel, Berta Bartoli Alabau, ..., Juan Carlos Garcia Pagan, Rémy Morello, Aurélie Plessier

## Correspondence

lebedel-l@chu-caen.fr (L. Lebedel).

## Graphical abstract



## Highlight

- High-risk prothrombotic factors are common in cases of recent non-cirrhotic portal vein thrombosis associated with local factors.
- High-risk prothrombotic factors are strongly associated with new thrombotic events.
- Stopping anticoagulation significantly increases the risk of rethrombosis when a high-risk prothrombotic factor is present.

## Impact and implications

In non-cirrhotic portal vein thrombosis (NCPVT) associated with local factors, systematic screening for prothrombotic factors is recommended, but the prevalence of the latter is not clearly established, and the risk of recurrent intra or extrasplanchnic thromboembolism is poorly described. Thus, interest in permanent anticoagulation remains. NCPVT associated with local factors is a matter of concern for hepatologists, gastroenterologists and digestive surgeons. Due to a lack of knowledge, practices are heterogeneous. Our findings highlight that systematic screening for prothrombotic factors in NCPVT is needed even when associated with local factors, as it may justify long-term anticoagulation for the prevention of new intra or extrasplanchnic thrombotic events in at least one-third of cases. The interest in long-term anticoagulation should be investigated prospectively in the absence of high-risk prothrombotic factors.

# Comparison of non-contrast abbreviated MRI and ultrasound as surveillance modalities for HCC

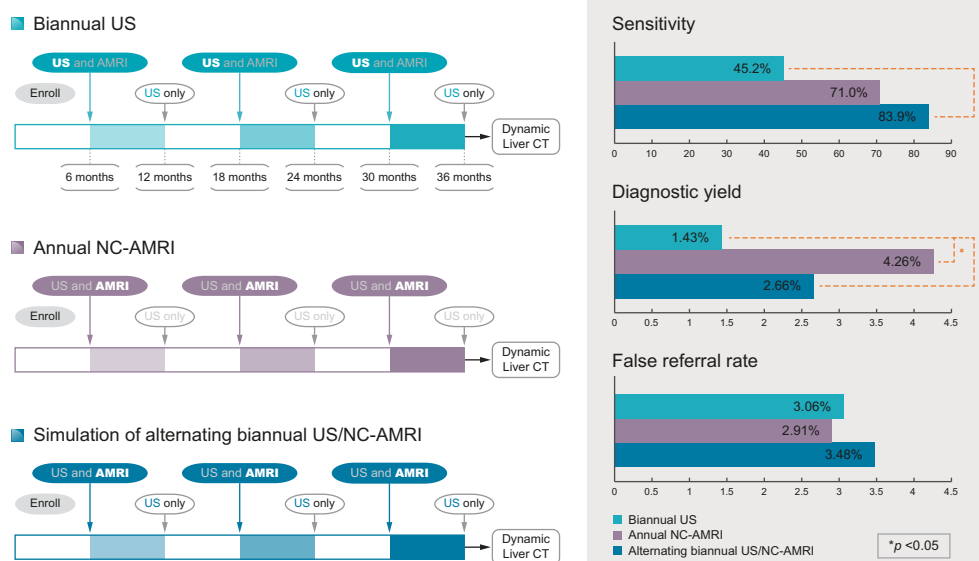
## Authors

Dong Hwan Kim, Jeong Hee Yoon, Moon Hyung Choi, ..., Kyung Ah Kim, Su Lim Lee, Joon-Il Choi

## Correspondence

dumkycji@gmail.com (J.-I. Choi).

## Graphical abstract



## Highlights

- Biannual US showed suboptimal performance for early-stage HCC detection.
- Annual NC-AMRI had significantly higher diagnostic yield than biannual US, without increasing the false referral rate.
- Alternating biannual US and NC-AMRI had significantly higher sensitivity than biannual US.

## Impact and implications

Current guidelines permit the use of magnetic resonance imaging (MRI) as a surveillance tool for hepatocellular carcinoma in patients in whom ultrasonography (US) is inadequate. However, the specific indications, imaging sequences, and intervals for MRI surveillance remain unclear. In our study, we found that annual non-contrast abbreviated MRI exhibited marginally higher sensitivity and significantly better diagnostic yield than biannual US in patients at high risk of hepatocellular carcinoma. Alternating US and non-contrast abbreviated MRI at 6-month intervals led to significantly improved sensitivity compared to biannual US, making it a potentially optimal surveillance strategy for high-risk patients.



# Transplant benefit-based offering of deceased donor livers in the United Kingdom

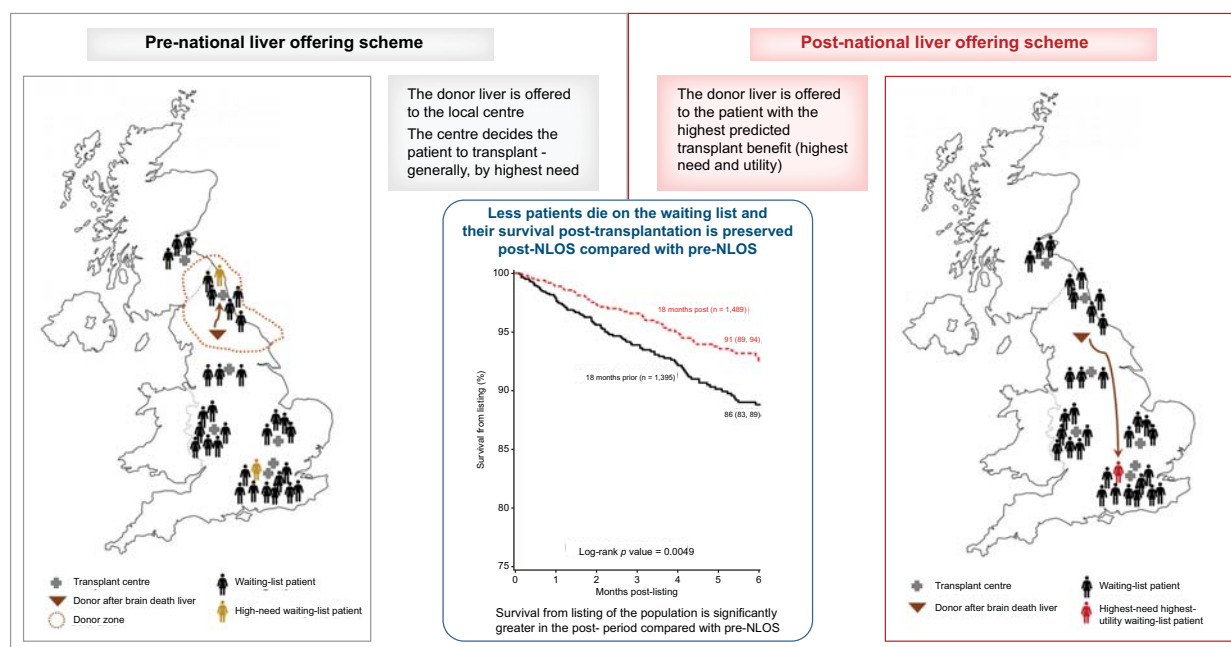
## Authors

Elisa Allen, Rhiannon Taylor, Alexander Gimson, Douglas Thorburn

## Correspondence

elisa.allen@nhsbt.nhs.uk (E. Allen).

## Graphical abstract



## Highlights

- In 2018 the UK introduced a benefit-based scheme to offer deceased donor livers.
- Benefit is predicted using a patient's need for, and the utility of, a transplant.
- The novelty of the scheme relates to the incorporation of predicted utility.
- The scheme reduces waiting-list mortality and preserves post-transplant outcomes.
- Life-year gain from transplantation is maximised for the waiting list population.

## Impact and implications

The National Liver Offering Scheme (NLOS) was introduced in the UK in 2018 to increase transparency of the deceased donor liver offering process, maximise the overall survival of the waiting list population, and improve equity of access to liver transplantation. To our knowledge, it is the first scheme that offers organs based on statistical prediction of transplant benefit: the transplant benefit score. The results are important to the transplant community – from healthcare practitioners to patients – and demonstrate that, in the first two years of NLOS offering, waiting list mortality fell while post-transplant survival was not negatively impacted, thus delivering on the scheme's objectives. The scheme continues to be monitored to ensure that the transplant benefit score remains up-to-date and that signals that suggest the possible disadvantage of some patients are investigated.

# EMC10 modulates hepatic ER stress and steatosis in an isoform-specific manner

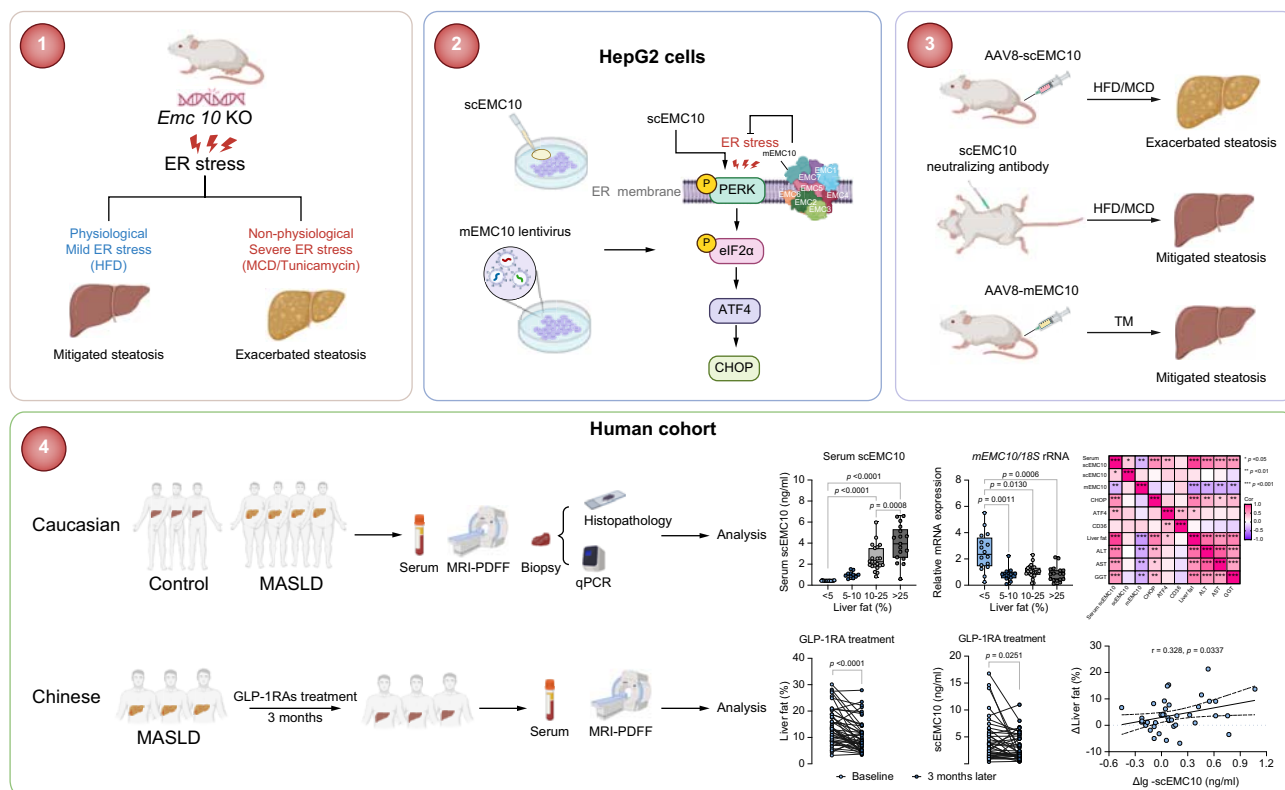
## Authors

Kuangyang Chen, Yahao Wang, Jia Yang, ..., Arne Dietrich, Matthias Blüher, Xuanchun Wang

## Correspondence

wangxch@fudan.edu.cn (X. Wang).

## Graphical abstract



## Highlights

- scEMC10 promotes, while mEMC10 suppresses, hepatic ER stress and steatosis in mice.
- scEMC10-neutralizing antibody ameliorates hepatosteatosis in mice with fatty liver.
- Serum scEMC10 positively, while hepatic mEMC10 negatively, correlates with MASLD in humans.
- GLP-1RAs decrease both serum scEMC10 and liver fat content in participants with MASLD.

## Impact and implications

We have shown the role of EMC10 in the regulation of energy homeostasis and obesity. In this study, we determine the distinct roles of the two isoforms of EMC10 in the regulation of hepatic endoplasmic reticulum stress and steatosis in mice, and report on the associations of the different EMC10 isoforms with metabolic dysfunction-associated steatotic liver disease in humans. Our findings delineate a novel regulatory axis for hepatosteatosis and identify EMC10 as a modulator of the PERK-eIF2 $\alpha$ -ATF4 signaling cascade that may be of broad physiological significance. Moreover, our pre-clinical and clinical studies provide evidence of the therapeutic potential of targeting scEMC10 in MASLD.